

Bonus Exercise: Quantifying BAP1 Exon 7

Now that you've successfully quantified exon 5, let's apply what you've learned to a new dataset: **BAP1 exon 7, library A**.

Learning Objectives

By completing this bonus exercise, you will:

- Apply the QUANTS workflow to a new dataset independently
- Create configuration files and samplesheets from scratch
- Understand how SGE parameters may differ between exons

Go to the tutorial directory

Before we begin, you need to go the tutorial directory:

```
cd /home/manager/MAVE_2025/course_data_2025/mave_data_analysis_1
```

Background: Why Exon 7?

Exon 7 of BAP1 is part of the UCH domain, just like exon 5, but spans a different functional region.

For this exercise: You'll quantify **Day 4 samples only** (3 biological replicates) to establish baseline library representation.

What You'll Need

Files Provided

In the `data/exon7/` directory, you'll find:

```
data/exon7/  
├── fastq/  
│   ├── 7_a_Day4_Rep1.fastq.gz  
│   ├── 7_a_Day4_Rep2.fastq.gz  
│   └── 7_a_Day4_Rep3.fastq.gz  
├── valiant_library_7a.csv  
└── nextflow.config
```

Files You Need to Create

You'll need to create:

1. **Samplesheet** (`samplesheet_7a.csv`)
2. **Configuration file** (`quants_7a.json`)

Task 1: Examine the Exon 7 Library

First, let's look at the ValiAnT library for exon 7 to understand what we're working with.

```
head data/exon7/valiant_library_7a.csv | column -t -s ','
```

```
wc -l data/exon7/valiant_library_7a.csv
```

Questions:

B1.1) How many variants are in the exon 7 library? (Remember to subtract 1 for the header)

Task 2: Create the Samplesheet

Create a samplesheet file called `samplesheet_7a.csv` in the `data/exon7/` directory. Complete samplesheet with details for all 3 exon 7 samples.

Template:

```
csv
sample,fastq_1,fastq_2
[SAMPLE_NAME],[PATH_TO_FASTQ],
```

Hints:

- Sample naming convention: `7_a_Day4_RepX` (where X = 1, 2, or 3)
- You have 3 samples (3 biological replicates from Day 4)
- FASTQ files are in `data/exon7/fastq/`
- Remember: single-end sequencing, so `fastq_2` is empty

Task 3: Determine Trimming Parameters

To create the configuration file, you need to figure out what adapters and primers to trim.

Step 1: Understand adapters and primers

Question:

B3.1) Why is the adapter the same for exon 7 as for exon 5?

Step 2: Identify the primers

The primers are **specific to each library** because they're used to amplify the HDR template.

How to find them:

For this exercise, we'll give you the primers:

```
Forward primer: GCTGTGGGAGCTGATGTGGG
Reverse primer: GAGCTGGGCTCAGGGCCCTCTGGTATGT
```

Why primers differ between exons:

- Primers are designed to amplify the specific HDR template
- Each exon is at a different genomic location
- Primers must match the flanking genomic sequence
- This is **biology-specific**, unlike adapters which are **technology-specific**

Task 3: Determine Read Modification Parameters

B4.1) Look at the exon 5 configuration. The read modification parameters were:

```
"append_start": "AATGATACGGCGACCACCGA",
"append_end": "TCGTATGCCGCTCTTCTGCTTG",
```

Are these likely to be the same for exon 7, or different?

Task 5: Create the Configuration File

Now create `quants_7a.json` in the `data/exon7/` directory.

*Note: this library only has 22 columns so `mseq` or the oligo sequences (python library convertor `-S`) is 22 for exon 7.

Starting template:

```
{
  "max_cpus": 4,
  "single_end": true,
  "input_type": "fastq",
  "raw_sequencing_qc": true,
  "adapter_trimming": "cutadapt",
  "adapter_trimming_qc": true,
  "adapter_cutadapt_options": "[YOUR_ADAPTER_SEQUENCE]",
  "primer_trimming": "cutadapt",
  "primer_trimming_qc": true,
  "primer_cutadapt_options": "[YOUR_PRIMER_SEQUENCES]",
  "read_modification": true,
  "append_start": "[YOUR_APPEND_START]",
  "append_end": "[YOUR_APPEND_END]",
  "append_quality": "?",
  "transform_library": true,
  "pyquest_library_converter_options": "-N 1 -S 22",
  "quantification": "pyquest"
}
```

Fill in the bracketed values using what you learned in Tasks 3 and 4.

Key points:

- Adapter: Same as exon 5 (Illumina infrastructure)
- Primers: Exon 7-specific (linked with ...)
- Append sequences: Same as exon 5 (synthesis method)

Task 6: Run QUANTS

Now run QUANTS on the exon 7 data!

Example command from exon5:

```
nextflow run /home/manager/QUANTS/main.nf \
-c ${PWD}/data/exon5/nextflow.config \
-params-file ${PWD}/data/exon5/quant_5a.json \
--input ${PWD}/data/exon5/samplesheet_5a.csv \
--oligo_library ${PWD}/data/exon5/valiant_library_5a.csv \
--outdir ${PWD}/results_5a
```

Update this command for exon7 and run it!

Expected runtime: ~10-15 minutes for 3 samples

Monitor progress: Watch for the process names and completion indicators just like you did for exon 5.

Task 7: Quality Control Assessment

Once QUANTS completes, assess the data quality using the tutorial information from exon5.

Summary

After completing this bonus exercise, you should:

- Be able to apply the QUANTS workflow independently to new datasets
- Understand which parameters are experiment-specific vs universal
- Know how to create configuration files from scratch
- Be comfortable with quality assessment across different exons

Key Insight: Most QUANTS parameters (adapters, append sequences) are consistent across experiments from the same lab/method. The main things that change are:

- Primers (library-specific)
- Library file (exon-specific)
- Sample names and file paths

Well done! You've now mastered SGE quantification and can confidently process new datasets independently.